

H524 Introduction to Biostatistics

Week 6: Power, Two-Sample Tests, and ANOVA

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Outline

- 1 Review from Week 5
- 2 Statistical Power (Chapter 11)
- 3 Two-Sample t-Tests: Independent Samples (Chapter 12)
- 4 Paired t-Tests (Chapter 12)
- 5 Analysis of Variance (ANOVA) (Chapter 12)
- 6 Summary and Key Formulas

Quick Review: Hypothesis Testing

Last week we covered:

- Null and alternative hypotheses
- Test statistics and p-values
- One-sample t-tests for means
- Type I error (false positive) and significance level α
- Rejection regions and critical values

This Week: Extending hypothesis testing

- Statistical power and Type II errors
- Comparing two independent samples
- Paired samples and dependent data
- Analysis of Variance (ANOVA) for multiple groups

Reading: Pagano & Gauvreau Chapters 11-12

Hypothesis Testing Errors

Two types of errors in hypothesis testing:

Decision	Truth	
	H_0 True	H_0 False
Reject H_0	Type I Error (False Positive) Probability = α	Correct Decision (True Positive) Probability = $1 - \beta$ (Power)
Fail to Reject H_0	Correct Decision (True Negative) Probability = $1 - \alpha$	Type II Error (False Negative) Probability = β

Key Definitions:

- α = Type I error rate (significance level, usually 0.05)
- β = Type II error rate (probability of missing real effect)
- **Power** = $1 - \beta$ (probability of detecting real effect)

What is Statistical Power?

Definition: Power is the probability of correctly rejecting H_0 when it is false.

$$\text{Power} = 1 - \beta = \Pr(\text{Reject } H_0 \mid H_0 \text{ is false})$$

Interpretation:

- Power = 0.80 means 80% chance of detecting a real effect
- If power is too low, you might miss important findings
- Typically want power ≥ 0.80 (80%)
- Some fields require power ≥ 0.90 (90%)

Example:

- Testing a new drug vs. placebo
- Drug truly reduces blood pressure by 10 mmHg
- Power = 0.80 means 80% chance our study will detect this effect
- 20% chance we'll conclude "no effect" (Type II error)

Factors Affecting Power

Power increases with:

1. Larger Sample Size (n)

- More data $\Rightarrow$ smaller standard error $\Rightarrow$ easier to detect effects
- Most important and controllable factor

2. Larger Effect Size ($|\mu - \mu_0|$ or $|\mu_1 - \mu_2|$)

- Bigger differences are easier to detect
- Not under researcher's control

3. Lower Variability (σ)

- Less noise $\Rightarrow$ clearer signal
- Can be reduced by careful measurement and study design

4. Higher Significance Level (α)

- Trade-off: higher α increases power but increases Type I error
- Usually keep $\alpha = 0.05$ fixed

Power Calculation Example

Scenario: Testing if a new drug lowers cholesterol

Given:

- Population mean cholesterol: $\mu_0 = 200$ mg/dL
- We want to detect a reduction to $\mu = 190$ mg/dL
- Known SD: $\sigma = 30$ mg/dL
- Sample size: $n = 25$
- Significance level: $\alpha = 0.05$ (two-sided)

Question: What is the power to detect this 10 mg/dL reduction?

Solution:

- 1 Standard error: $SE = 30/\sqrt{25} = 6$ mg/dL
- 2 Critical value: $z_{0.975} = 1.96$
- 3 Rejection region: Reject if $\bar{x} < 200 - 1.96(6) = 188.24$ or $\bar{x} > 211.76$
- 4 Under true $\mu = 190$: Calculate $\Pr(\bar{X} < 188.24 \text{ or } \bar{X} > 211.76 \mid \mu = 190)$

Calculate power for the cholesterol example:

```
# Parameters
mu0 <- 200; mu1 <- 190; sigma <- 30; n <- 25; alpha <- 0.05

# Standard error
se <- sigma / sqrt(n) # 6

# Critical values (two-sided)
z_crit <- qnorm(1 - alpha/2) # 1.96
lower_crit <- mu0 - z_crit * se # 188.24
upper_crit <- mu0 + z_crit * se # 211.76

# Power: P(reject H0 | mu = 190)
power_lower <- pnorm(lower_crit, mean=mu1, sd=se)
power_upper <- pnorm(upper_crit, mean=mu1, sd=se,
                     lower.tail=FALSE)
power <- power_lower + power_upper
cat("Power:", round(power, 3)) # 0.385
```

Power Calculation Results

Result: Power = 0.385 (38.5%)

Interpretation:

- Only 38.5% chance of detecting a 10 mg/dL reduction with $n = 25$
- 61.5% chance of Type II error (missing the effect)
- This is too low! We want power ≥ 0.80

Solution: Increase sample size

For power = 0.80, we need approximately $n = 73$ subjects

General lesson:

- Always calculate power **before** conducting study
- Helps determine necessary sample size
- Avoids wasting resources on underpowered studies
- Part of good study design and grant proposals

Sample Size Calculation in R

Using `power.t.test()` function:

```
# Calculate required sample size for power = 0.80
power.t.test(delta = 10,      # Effect size ( $|\mu_1 - \mu_0|$ )
              sd = 30,        # Standard deviation
              sig.level = 0.05, # Significance level
              power = 0.80,    # Desired power
              type = "one.sample")
```

Output:

```
#      One-sample t test power calculation
#              n = 72.58
#           delta = 10
#              sd = 30
#    sig.level = 0.05
#         power = 0.8
# Need n = 73 subjects
```

Sample Size Calculation in R (continued)

Calculate power for given sample size:

```
# Can also calculate power for given n
power.t.test(n = 25, delta = 10, sd = 30,
             sig.level = 0.05, type = "one.sample")
```

Output:

```
#      One-sample t test power calculation
#              n = 25
#           delta = 10
#              sd = 30
#      sig.level = 0.05
#           power = 0.360
# Power = 0.360 (slightly lower than z-based calculation)
```

Visualizing how power changes with sample size:

[Power increases non-linearly with sample size]

Key observations:

- Power increases rapidly initially
- Diminishing returns for very large n
- To double power from 0.40 to 0.80, need $\sim 4\times$ sample size
- To cut standard error in half, need $4\times$ sample size

Practical implications:

- Small studies ($n < 30$) often severely underpowered
- “Pilot studies” rarely have power to detect effects
- Need careful planning and power analysis before data collection
- Consider effect size: smaller effects need larger samples

Comparing Two Independent Groups

Common research questions:

- Does treatment A work better than treatment B?
- Do men and women differ in average blood pressure?
- Is mean cholesterol higher in smokers vs. non-smokers?

Two-sample problem:

- Two independent groups (treatment/control, male/female, etc.)
- Want to compare population means: μ_1 vs. μ_2
- Observe sample means: $\bar{x}_1$ and $\bar{x}_2$

Hypotheses:

$$H_0 : \mu_1 = \mu_2 \quad (\text{or equivalently: } \mu_1 - \mu_2 = 0)$$

$$H_A : \mu_1 \neq \mu_2 \quad (\text{two-sided})$$

Or one-sided: $H_A : \mu_1 > \mu_2$ or $H_A : \mu_1 < \mu_2$

Two-Sample t-Test: Assumptions

Assumptions for validity:

1. Independence

- Observations within each group are independent
- The two groups are independent of each other
- Violated by paired data (covered later)

2. Normality

- Data in each group approximately normal
- Or large sample sizes ($n_1, n_2 \geq 30$) for CLT
- Check with histograms, Q-Q plots

3. Equal Variances (for pooled t-test)

- $\sigma_1^2 = \sigma_2^2$ (homogeneity of variance)
- Can test with F-test or Levene's test
- If violated, use Welch's t-test (does not assume equal variances)

Pooled Two-Sample t-Test

When variances are equal: $\sigma_1^2 = \sigma_2^2 = \sigma^2$

Test statistic:

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

where the **pooled standard deviation** is:

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

Distribution: Under H_0 , $t \sim t_{n_1+n_2-2}$

Degrees of freedom: $df = n_1 + n_2 - 2$

Decision rule:

- Calculate p-value: $p = 2 \times \Pr(T > |t| \mid H_0)$ for two-sided
- Reject H_0 if $p < \alpha$

Example: Tumor Growth Study

Study: Effect of new drug on tumor growth (mm/month)

Control	Treatment
7	4
10	6
9	10
8	8
7	5
6	3
8	10
9	8
12	8
13	10

Question: Does the treatment reduce tumor growth?

Hypotheses:

- $H_0 : \mu_C = \mu_T$ (no difference)
- $H_A : \mu_C > \mu_T$ (treatment reduces growth)

Two-Sample t-Test in R (Part 1 - Setup)

```
# Enter data
control <- c(7, 10, 9, 8, 7, 6, 8, 9, 12, 13)
treatment <- c(4, 6, 10, 8, 5, 3, 10, 8, 8, 10)

# Summary statistics
mean(control)      # 8.9 mm
mean(treatment)    # 7.2 mm
sd(control)        # 2.234 mm
sd(treatment)      # 2.573 mm

# Pooled two-sample t-test (equal variances)
t.test(control, treatment,
        var.equal = TRUE,      # Assume equal variances
        alternative = "greater") # One-sided: control > treatment
```

Two-Sample t-Test in R (Part 2 - Results)

```
# Output from t.test():  
# t = 1.578, df = 18, p-value = 0.066  
# mean of x = 8.9, mean of y = 7.2  
# 95% CI: (-0.182, Inf)  
  
# Conclusion:  $p = 0.066 > 0.05$ , fail to reject  $H_0$   
# Not enough evidence that treatment reduces growth
```

Welch's t-Test (Unequal Variances)

When variances are unequal: $\sigma_1^2 \neq \sigma_2^2$

Test statistic:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Degrees of freedom (Welch-Satterthwaite):

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1-1} + \frac{(s_2^2/n_2)^2}{n_2-1}}$$

In R: Just use `t.test()` with `var.equal = FALSE` (default!)

When to use:

- Always safer to use Welch's test
- More robust, similar power when variances equal
- Default in R's `t.test()` function

Confidence Interval for Difference in Means

95% CI for $\mu_1 - \mu_2$ (pooled):

$$(\bar{x}_1 - \bar{x}_2) \pm t_{0.975, df} \times s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

where $df = n_1 + n_2 - 2$

Tumor example:

- $\bar{x}_C - \bar{x}_T = 8.9 - 7.2 = 1.7$ mm
- $s_p = 2.409$ mm
- $t_{0.975, 18} = 2.101$
- $SE = 2.409 \sqrt{\frac{1}{10} + \frac{1}{10}} = 1.078$ mm
- 95% CI: $1.7 \pm 2.101 \times 1.078 = 1.7 \pm 2.26$
- **95% CI: (-0.56, 3.96) mm**

Interpretation: We are 95% confident the true difference is between -0.56 and 3.96 mm. Since interval includes 0, no significant difference at $\alpha = 0.05$.

Paired vs. Independent Samples

Paired (dependent) samples:

- Same subjects measured twice (before/after)
- Matched pairs (twins, siblings, matched cases)
- Left vs. right measurements on same subject

Examples:

- Blood pressure before and after medication
- Weight before and after diet intervention
- Pain score before and after treatment
- Test scores with two different methods on same students

Why pairing matters:

- Removes between-subject variability
- More powerful than two-sample t-test
- Controls for confounding variables
- **Mistake:** Using two-sample test on paired data loses power

Paired t-Test Procedure

Key idea: Analyze the *differences*

For each pair i :

$$d_i = x_{i,\text{before}} - x_{i,\text{after}}$$

or $d_i = x_{i,1} - x_{i,2}$

Then: Perform one-sample t-test on the differences

Hypotheses:

$$H_0 : \mu_d = 0 \quad (\text{no change})$$

$$H_A : \mu_d \neq 0 \quad (\text{two-sided})$$

Test statistic:

$$t = \frac{\bar{d} - 0}{s_d / \sqrt{n}}$$

where $\bar{d}$ = mean difference, s_d = SD of differences, n = number of pairs

Distribution: Under H_0 , $t \sim t_{n-1}$

Example: Blood Pressure Medication

Study: 10 patients, blood pressure measured before and after medication

Patient	Before	After	Difference (d_i)
1	145	138	7
2	150	142	8
3	148	145	3
4	142	138	4
5	140	135	5
6	146	140	6
7	149	142	7
8	143	136	7
9	147	141	6
10	144	138	6
Mean	145.4	139.5	5.9
SD	3.13	3.10	1.52

Question: Does medication significantly reduce blood pressure?

Paired t-Test in R (Part 1 - Setup)

```
# Enter data
before <- c(145, 150, 148, 142, 140, 146, 149, 143, 147, 144)
after <- c(138, 142, 145, 138, 135, 140, 142, 136, 141, 138)

# Calculate differences
differences <- before - after
mean(differences)  # 5.9 mmHg
sd(differences)    # 1.52 mmHg

# Paired t-test (Method 1: on differences)
t.test(differences, mu = 0, alternative = "greater")

# Paired t-test (Method 2: paired argument)
t.test(before, after, paired = TRUE, alternative = "greater")
```

Paired t-Test in R (Part 2 - Results)

```
# Output from paired t-test:  
# t = 12.24, df = 9, p-value = 0.00000032  
# mean of differences = 5.9  
# 95% CI: (5.05, Inf)  
  
# Conclusion:  $p < 0.001$ , strong evidence that  
# medication reduces blood pressure by 5.9 mmHg on average
```

Paired vs. Independent t-Test Comparison

What if we (incorrectly) used two-sample test?

Using two-sample t-test: $p = 0.115$ (not significant!)

Using paired t-test: $p = 0.0001$ (highly significant!)

Why such a big difference?

- Pairing removes between-subject variability
- Standard error is much smaller: $SE = 2.78/\sqrt{10} = 0.88$
- Two-sample SE: $\sqrt{8.23^2/10 + 6.86^2/10} = 3.25$ (nearly 4x larger!)
- Pairing gives much more power

Lesson:

- Always use paired test for paired data
- Pairing is a powerful design strategy
- Can dramatically reduce sample size needed

Comparing More Than Two Groups

Limitation of t-tests: Only compare two groups at a time

What if we have 3+ groups?

- Compare 3 drug doses: low, medium, high
- Compare 4 treatment methods
- Compare outcomes across multiple hospitals

Naive approach: Multiple t-tests

- Compare groups 1 vs. 2, 1 vs. 3, 2 vs. 3, etc.
- For k groups: $\binom{k}{2} = \frac{k(k-1)}{2}$ comparisons
- For $k = 3$: 3 tests; for $k = 4$: 6 tests; for $k = 5$: 10 tests

Problem: Multiple testing inflation

- Each test has Type I error rate $\alpha = 0.05$
- Overall Type I error rate much higher!
- For 3 independent tests: $1 - (1 - 0.05)^3 = 0.143$ (14.3%)
- Need a method that controls overall error rate

Analysis of Variance (ANOVA)

Solution: One omnibus test for all groups

Hypotheses:

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_k \quad (\text{all means equal})$$

H_A : At least one mean differs

Key idea: Compare variation *between* groups to variation *within* groups

- **Between-group:** How different are the group means?
- **Within-group:** How variable within each group?

If H_0 is true:

- All groups sampled from same population
- Between-group $\approx$ within-group variation
- Differences just due to chance

If H_A is true:

- Between-group $>$ within-group variation
- Group means genuinely different

ANOVA: Partitioning Variability (Part 1)

Total variability = Between-group + Within-group

Sum of Squares decomposition:

$$SS_{\text{Total}} = SS_{\text{Between}} + SS_{\text{Within}}$$

Total Sum of Squares:

$$SS_{\text{Total}} = \sum_{i=1}^k \sum_{j=1}^{n_i} (x_{ij} - \bar{x})^2$$

where x_{ij} = observation j in group i , $\bar{x}$ = overall mean

ANOVA: Partitioning Variability (Part 2)

Between-Group Sum of Squares:

$$SS_{\text{Between}} = \sum_{i=1}^k n_i (\bar{x}_i - \bar{x})^2$$

where $\bar{x}_i$ = mean of group i

Within-Group Sum of Squares:

$$SS_{\text{Within}} = \sum_{i=1}^k \sum_{j=1}^{n_i} (x_{ij} - \bar{x}_i)^2$$

Measures variation of observations around their group means

ANOVA: Mean Squares and F-Statistic

Mean Squares = Sum of Squares / degrees of freedom

Between-Group Mean Square:

$$MS_{\text{Between}} = \frac{SS_{\text{Between}}}{k - 1}$$

Within-Group Mean Square (pooled variance):

$$MS_{\text{Within}} = \frac{SS_{\text{Within}}}{n - k}$$

where k = number of groups, n = total sample size

F-Statistic:

$$F = \frac{MS_{\text{Between}}}{MS_{\text{Within}}}$$

Distribution: Under H_0 , $F \sim F_{k-1, n-k}$

Decision rule:

- Large $F \Rightarrow$ evidence against H_0
- Calculate p-value: $p = \Pr(F_{k-1, n-k} > F_{\text{obs}})$
- Reject H_0 if $p < \alpha$

ANOVA Table

Standard ANOVA output format:

Source	SS	df	MS	F
Between Groups	SS_{Between}	$k - 1$	MS_{Between}	F
Within Groups	SS_{Within}	$n - k$	MS_{Within}	
Total	SS_{Total}	$n - 1$		

Interpretation:

- Large F-statistic $\Rightarrow$ between-group variation is large relative to within-group
- Small p-value $\Rightarrow$ reject H_0 , conclude at least one mean differs
- If reject H_0 : Follow up with post-hoc tests to see which groups differ

Note: ANOVA tells us there's a difference somewhere, but not where!

ANOVA Assumptions

For valid ANOVA results:

1. Independence

- Observations within and between groups are independent
- Random sampling or random assignment

2. Normality

- Data within each group approximately normal
- Robust to violations with large samples (CLT)
- Check with histograms, Q-Q plots

3. Equal Variances (Homogeneity)

- $\sigma_1^2 = \sigma_2^2 = \dots = \sigma_k^2$
- Check with Levene's test or Bartlett's test
- Visually: boxplots should have similar spreads
- If violated: Use Welch's ANOVA or transformation

4. Balanced Design (preferred but not required)

- Equal sample sizes in each group increases power

Example: Drug Dose Study

Study: Effect of 3 drug doses on cholesterol reduction (mg/dL)

Low Dose	Medium Dose	High Dose
10	15	25
12	18	28
8	16	30
11	20	26
9	17	27
Mean: 10.0 SD: 1.58	Mean: 17.2 SD: 1.92	Mean: 27.2 SD: 1.92

Question: Is there a significant difference in cholesterol reduction across doses?

Hypotheses:

- $H_0 : \mu_{\text{low}} = \mu_{\text{med}} = \mu_{\text{high}}$
- $H_A : \text{At least one mean differs}$

ANOVA in R (Part 1 - Setup)

```
# Enter data
low <- c(10, 12, 8, 11, 9)
medium <- c(15, 18, 16, 20, 17)
high <- c(25, 28, 30, 26, 27)

# Create data frame
cholesterol <- c(low, medium, high)
dose <- factor(rep(c("Low", "Medium", "High"), each=5),
               levels=c("Low", "Medium", "High"))
data <- data.frame(cholesterol, dose)

# Perform ANOVA
model <- aov(cholesterol ~ dose, data=data)
summary(model)
```

ANOVA in R (Part 2 - Results)

```
# Output from summary(model):  
#           Df Sum Sq Mean Sq F value Pr(>F)  
# dose      2  746.1   373.1   113.1 1.64e-08 ***  
# Residuals 12   39.6     3.3  
  
# Conclusion:  $F(2,12) = 113.1$ ,  $p < 0.001$   
# Strong evidence that at least one dose differs
```

After significant ANOVA: Which groups differ?

Problem: Multiple pairwise comparisons inflates Type I error

Solution: Post-hoc tests that control family-wise error rate

Common post-hoc tests:

1. Tukey's HSD (Honestly Significant Difference)

- Most common and powerful
- Controls overall Type I error at α
- All pairwise comparisons

2. Bonferroni Correction

- Simple but conservative
- Use $\alpha^* = \alpha/m$ for each test (m = number of comparisons)
- For 3 groups: $\alpha^* = 0.05/3 = 0.0167$

3. Scheffe's Method

- Most conservative
- Allows any contrasts, not just pairwise

Post-Hoc Tests in R (Part 1 - Tukey's HSD)

```
# Tukey's HSD test
TukeyHSD(model)

# Output:
#   Tukey multiple comparisons of means
#     95% family-wise confidence level
#
# Fit: aov(formula = cholesterol ~ dose, data = data)
#
# $dose
#               diff          lwr          upr          p adj
# Medium-Low      7.2000    4.413    10.0870    0.000032
# High-Low        17.2000   14.413    19.9870    0.000000
# High-Medium     10.0000    7.213    12.7870    0.000001

# All pairwise differences are significant
# High dose reduces cholesterol significantly more than
# medium dose, which reduces more than low dose
```

Post-Hoc Tests in R (Part 2 - Alternative)

```
# Alternative: Pairwise t-tests with Bonferroni correction
pairwise.t.test(data$cholesterol, data$dose,
                 p.adjust.method = "bonferroni")

# Output:
#           Low      Medium
# Medium 3.2e-5    -
# High   < 2e-16 9.5e-7

# Results are similar to Tukey's HSD
# All pairwise differences remain highly significant
```

ANOVA vs. Multiple t-Tests

Why ANOVA instead of multiple t-tests?

Advantages of ANOVA:

1 Controls Type I error rate

- One test at $\alpha = 0.05$ level
- Multiple t-tests inflate error rate

2 More powerful

- Pools variance estimate across all groups
- More degrees of freedom for error

3 Can test complex hypotheses

- Test overall effect of categorical variable
- Can include multiple factors (two-way ANOVA)

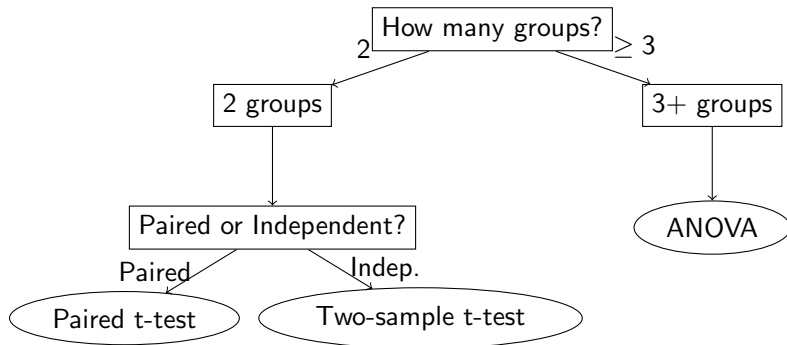
4 Provides comprehensive summary

- Single test answers overall question
- Follow up only if significant

When to use t-test: Only 2 groups to compare

Choosing the Right Test

Decision tree:



Additional considerations:

- Equal variances? Use pooled vs. Welch's t-test
- Normal data? If not, non-parametric tests (Week 7)
- Small samples + non-normal? Use caution

Key Formulas: Power and Sample Size

Type I and Type II Errors:

- $\alpha = P(\text{Reject } H_0 \mid H_0 \text{ true}) = \text{Type I error}$
- $\beta = P(\text{Fail to reject } H_0 \mid H_0 \text{ false}) = \text{Type II error}$
- $\text{Power} = 1 - \beta = P(\text{Reject } H_0 \mid H_0 \text{ false})$

Power depends on:

- Sample size n (larger $\Rightarrow$ more power)
- Effect size $|\mu_1 - \mu_0|$ (larger $\Rightarrow$ more power)
- Variability σ (smaller $\Rightarrow$ more power)
- Significance level α (larger $\Rightarrow$ more power, but more Type I error)

In R:

- `power.t.test()` for one-sample and two-sample
- Specify any 4 of: `n`, `delta`, `sd`, `sig.level`, `power`
- Function solves for the missing parameter

Key Formulas: Two-Sample t-Test

Pooled two-sample t-test (equal variances):

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \quad \text{where} \quad s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

Distribution: $t \sim t_{n_1+n_2-2}$ under $H_0 : \mu_1 = \mu_2$

Confidence interval for $\mu_1 - \mu_2$:

$$(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2, n_1+n_2-2} \times s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

Welch's t-test (unequal variances):

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

with complicated df formula (R calculates automatically)

Key Formulas: Paired t-Test

Paired t-test:

For differences $d_i = x_{i,1} - x_{i,2}$:

$$t = \frac{\bar{d}}{s_d / \sqrt{n}}$$

where $\bar{d}$ = mean difference, s_d = SD of differences

Distribution: $t \sim t_{n-1}$ under $H_0 : \mu_d = 0$

Confidence interval for μ_d :

$$\bar{d} \pm t_{\alpha/2, n-1} \times \frac{s_d}{\sqrt{n}}$$

When to use:

- Before/after measurements on same subjects
- Matched pairs (twins, siblings, matched controls)
- Left/right measurements
- Any dependent samples

Key Formulas: ANOVA

F-statistic:

$$F = \frac{MS_{\text{Between}}}{MS_{\text{Within}}} = \frac{SS_{\text{Between}}/(k-1)}{SS_{\text{Within}}/(n-k)}$$

Distribution: $F \sim F_{k-1, n-k}$ under $H_0 : \mu_1 = \dots = \mu_k$

Sum of squares:

$$SS_{\text{Total}} = \sum_i \sum_j (x_{ij} - \bar{x})^2$$

$$SS_{\text{Between}} = \sum_i n_i (\bar{x}_i - \bar{x})^2$$

$$SS_{\text{Within}} = \sum_i \sum_j (x_{ij} - \bar{x}_i)^2$$

Relationship: $SS_{\text{Total}} = SS_{\text{Between}} + SS_{\text{Within}}$

Post-hoc: If F-test significant, use Tukey's HSD or Bonferroni to identify which groups differ

Summary: This Week

Today we covered:

● Chapter 11: Power

- Type I and Type II errors
- Statistical power and factors affecting it
- Sample size calculations

● Chapter 12: Two-Sample Tests

- Independent two-sample t-test (pooled and Welch's)
- Paired t-test for dependent samples
- Confidence intervals for differences

● Chapter 12: ANOVA

- Comparing 3+ groups simultaneously
- F-test and ANOVA table
- Post-hoc tests (Tukey, Bonferroni)

Next Week: Non-parametric tests (Chapter 13)

Reading: Chapter 13 (Wilcoxon, Kruskal-Wallis, Sign test)

Thank you!

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